

Finite fields

E.g., $\frac{\mathbb{Z}_2[x]}{\langle x^2+x+1 \rangle}$

If $p(x) \in F[x]$, F a field, $p(x)$ irreducible
 $K = \frac{F[x]}{\langle p(x) \rangle}$ is a field with degree of
 ext'n ($i.e.$ $\dim_F K$) is $\deg p(x)$

Q find all quadratic
irr. poly in $\mathbb{Z}_2[x]$.

($\mathbb{Z}_2$ is also denoted by $\mathbb{F}_2$)

$$\mathbb{F}_4 := \frac{\mathbb{Z}_2[x]}{\langle x^2+x+1 \rangle} \text{ or } \mathbb{F}_4$$

$$\parallel$$

$$\{ \bar{0}, \bar{1}, \bar{x}, \bar{x}+1 \}$$

& Verify $\bar{x}+1 = \bar{x}^2$

$$(\text{because } \bar{x}^2 + \bar{x} + \bar{1} = \bar{0})$$

$$\bar{x}^2 = -\bar{x} - \bar{1} = \bar{x} + \bar{1}$$

People write $\mathbb{F}_4 = \{0, 1, \alpha, \alpha+1\}$
 where $\alpha^2 = 1 + \alpha$.

$$\text{Observe } (\alpha+1)^2 = \alpha^2 + 1 = \alpha + 1 + 1 = \alpha$$

Find a cubic irr. polynomial
 over $\mathbb{F}_2$.

$$\begin{array}{cccc} x^3 & x^2 & x & 1 \\ \hline x^3+x+1 & & & \\ x^3+x^2+1 & & & \end{array}$$

is irr.

$$\mathbb{F}_8 = \frac{\mathbb{F}_2[x]}{\langle x^3+x+1 \rangle} = \{ \bar{0}, \bar{1}, \bar{x}, \bar{x}+1, \bar{x}^2, \bar{x}^2+1, \bar{x}^2+x, \bar{x}^2+x+1 \}$$

Recall $\dim_{\mathbb{F}_2} \mathbb{F}_8 = 3$ ($= \deg(x^3+x+1)$) so that $|\mathbb{F}_8| = |\mathbb{F}_2|^3 = 2^3 = 8$

Over $\mathbb{F}_p$, p is a prime, a quadratic irr. poly exists!

A quadratic ~~poly~~ monic polynomial over $\mathbb{F}_p$ is of the form

$$x^2 + ax + b \text{ which cannot be written as } (x-\alpha)(x-\beta)$$

for $\alpha, \beta \in \mathbb{F}_p$.

The number of irreducible quadratic polynomials of
 degree 2 is the number of poly of the form x^2+ax+b
 — " " " " " " " " $(x-\alpha)(x-\beta)$

$$= p^2 - \frac{p^2+p}{2} = \frac{p^2-p}{2} = \boxed{\frac{p(p-1)}{2}}$$

$$\left[\text{For } p=2, \quad 2^2-1=1 \right] \quad \frac{2^2-2}{2} = 2 \times \frac{1}{2} = 1$$

$$\left\{ \begin{array}{l} (0,0) \quad (0,1) \quad \dots \quad (0,p-1) \\ (1,0) \quad (1,1) \quad \dots \quad (1,p-1) \\ \vdots \\ (p-1,0) \quad (p-1,1) \quad \dots \quad (p-1,p-1) \end{array} \right\}$$

$$\begin{aligned} & \frac{p^2-p}{2} + p \\ &= \frac{p^2+p}{2} < p^2 \end{aligned}$$

Question: Find the number of cubic irreducible polynomials over $\mathbb{F}_p$.

Thm (*) If $f(x) \in \mathbb{F}[x]$, then $\exists$ an extⁿ K of $\mathbb{F}$ s.t. $f(x)$ gets factored into linear factors.

We have seen

Ex: If $f(x) \in \mathbb{F}[x]$ is irr. then $K = \frac{\mathbb{F}[x]}{\langle f(x) \rangle}$ has a root of $f(x) \in \mathbb{F}[x] \subset K[x]$

In fact, $\bar{y} = y + \langle f(x) \rangle$ is a root of $f(x)$

Recall if $f(x) = a_0 + a_1 x + \dots + a_{n-1} x^{n-1} + x^n$

then in $K[x]$, $f(x) = \bar{a}_0 + \bar{a}_1 x + \dots + \bar{a}_{n-1} x^{n-1} + x^n$.

$$\left\{ \begin{array}{l} \mathbb{F} \hookrightarrow K = \frac{\mathbb{F}[x]}{\langle f(x) \rangle} \\ a \mapsto \bar{a} \\ \parallel \\ \langle f(x) \rangle + a \end{array} \right.$$

To prove $\mathbb{R}^n(x)$ factorize $f(x)$ into irr. poly.

Say $f(x) = g_1(x) h(x)$ $g_1(x)$ irr.

If $\deg g_1(x) \neq 1$, then find an extⁿ where $g_1(x)$ has a root & ~~hence~~ $f(x)$ has a root.

This process to be performed finitely many times to factor $f(x)$ into linear ~~poly~~ polynomials.

A minimal extension of $\mathbb{F}$ where $f(x)$ gets factored into linear factors is called a splitting field of $f(x)$ over $\mathbb{F}$.
Splitting field of $f(x)$ over $\mathbb{F}$ is unique upto isomorphism.

Consider $l(x) = x^{p^n} - x \in \mathbb{F}_p[x]$

Let $\mathbb{F}_{p^n}$ be the splitting field of $l(x)$.

Let $\alpha, \beta \in \mathbb{F}_{p^n}$ be s.t.

$$\frac{l(\alpha) = 0 = l(\beta)}{\alpha^{p^n} = \alpha}$$

$$\begin{aligned} l(\alpha + \beta) &= (\alpha + \beta)^{p^n} - (\alpha + \beta) \\ &= \alpha^{p^n} + \beta^{p^n} - (\alpha + \beta) \\ &= (\alpha^{p^n} - \alpha) + (\beta^{p^n} - \beta) \\ &= 0 + 0 \\ &= 0 \end{aligned}$$

Since $\binom{p^n}{j}$ is divisible by p for $\alpha^j < p^n$
so that $\binom{p^n}{j} = 0$ in $\mathbb{F}_p$

$$l(\alpha^{-1}) = (\alpha^{-1})^{p^n} - \alpha^{-1} = (\alpha^{p^n})^{-1} - \alpha^{-1}$$

$$= \alpha^{-1} - \alpha^{-1} = 0.$$

So if α, β are roots of $f(x)$ then
 so are $\alpha + \beta$ & α^{-1} & $\alpha\beta$.

~~S/L $\alpha\beta$~~

$$0^{p^n} - 0 = 0$$

$$1^{p^n} - 1 = 0$$

So the roots of $x^{p^n} - x = f(x)$ form a field.

Therefore the splitting field of $x^{p^n} - x$ over $\mathbb{F}_p$ is
 ~~$\mathbb{F}_p$~~ $\mathbb{F}_{p^n}$ consists of roots of $x^{p^n} - x$.

$$\mathbb{F}_3 \xrightarrow{f(x) = x^2 + 1} \mathbb{F}_3[x] \xrightarrow{\frac{x^3 - x}{\langle x^2 + 1 \rangle}} \frac{\mathbb{F}_3[x]}{\langle x^2 + 1 \rangle}$$

$$\overline{\mathbb{F}_3} = \frac{\mathbb{F}[x]}{\langle ? \rangle}$$

$$y^2 = x^3 + ax + b \quad \text{over } \mathbb{F}^2$$

$$y^2 = x^3 + ax^2 + bx^2 \quad \text{over } \mathbb{F}^2$$